Computer Vision
How Machines "SEE"

Why machines need VISION ?

- The Internet of Things, or IoT, refers to the billions of physical devices around the world that are now connected to the internet, all collecting and sharing data.

IoT and Vision

Confused ?

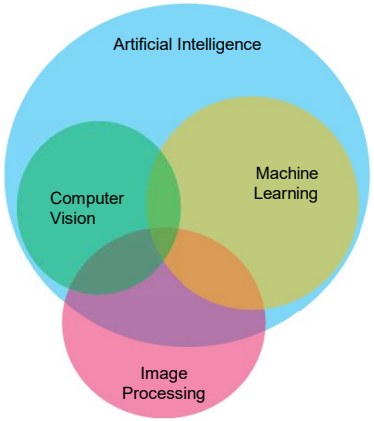
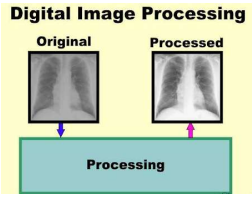


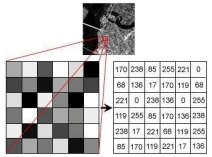
Image Processing

- Digital image processing is the use of a digital computer to process digital images through an algorithm.
- As a subcategory or field of digital signal processing



Digital Image

[0,0]	[0,1]	[0,2]	[0,3]	[0,4]
[1,0]	[1,1]	[1,2]	[1,3]	[1,4]
[2,0]	[2,1]	[2,2]	[2,3]	[2,4]
[3,0]	[3,1]	[3,2]	[3,3]	[3,4]
[4,0]	[4,1]	[4,2]	[4,3]	[4,4]



- **Rectangular grid of pixels- 5x5 grid**

If we are using 1 bit per cell, how many bits are needed to present the picture ?

- **What is a pixel?**

Point/Cell in the image that contains color data

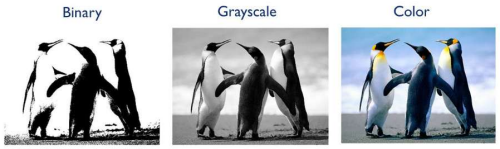
Each pixel is made up of bits

- **Resolution:** Details contained in an image

Defined by the number of pixels

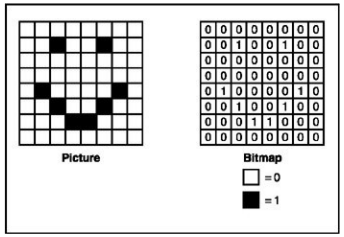
Image Types

- Binary
- Gray Scale
- Color



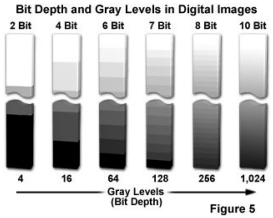
Binary Images

- Each pixel is represented using a bit (1 or 0)

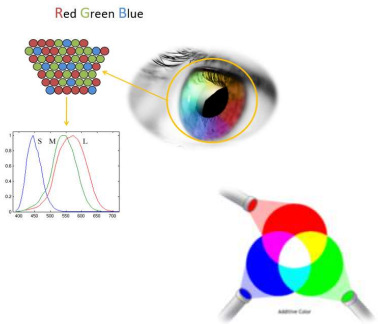


Grayscale Images

- Each pixel is represented using n bit
- So it can have 2^n gray levels



Color Images



Red	Green	Blue	Color
•255 •255	•255 •255	•255 •0	•White •Yellow

Image Processing: Filtering

Enhancing an image's quality for human viewing, e.g., in medical imaging or in telescopic views of space

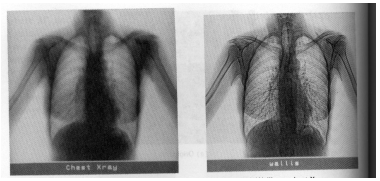


Image Processing: Watermarking

Information hiding

Protecting copyright

Image Processing: Compression

- Lossless – No loss in quality: gif, tiff
- Lossy – Original image cannot be reconstructed: jpeg

100% 53%

Image Processing: Morphology

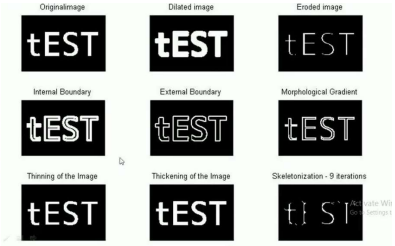


Image Processing: Edge Detection

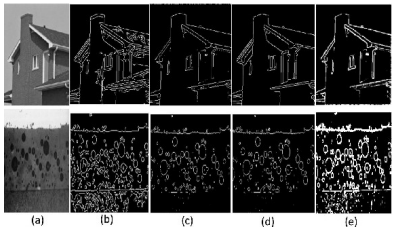
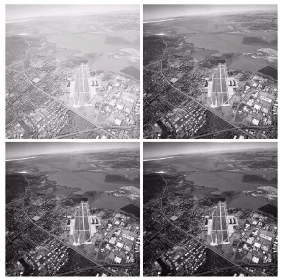


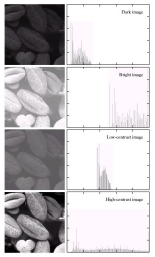
Image Enhancement



a b
c d

FIGURE 3.9 (a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with c = 1 and γ = 3.0, 4.0, and 5.0, respectively. (Original image for this example courtesy of NASA.)

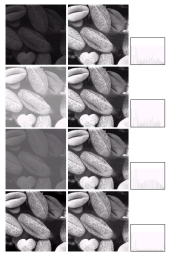
Image Histograms



a b

FIGURE 3.15 Four basic image types: dark, light, low contrast, high contrast, and their corresponding histograms. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

Histogram Equalization



a b c

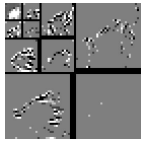
FIGURE 3.17 (a) Images from Fig. 3.15. (b) Results of histogram equalization. (c) Corresponding histograms.

Computer Vision

Image Transformation

- Transforming image can make it easier to analyze

Image Restoration



Wavelet transform of image

Image Reconstruction

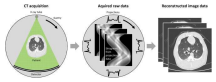
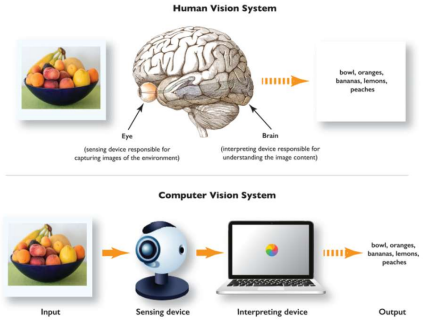


Image Super resolution

Computer Vision

- A field of AI that trains computers to understand and interpret visual world



Computer Vision Vs Image Processing

- Image Processing
- Research area within signal processing
- Focus on syntax,low level features

image image

- Computer Vision
- Research area within computer science/AI
- Focus on semantics, symbolic or geometric descriptions

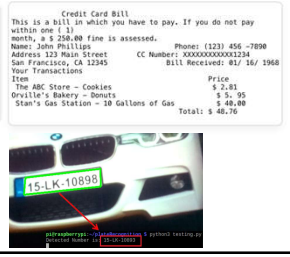
image

Objects,
Semantics,
Captions,
…….
…….

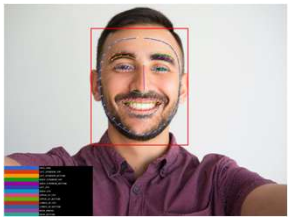
Optical character recognition (OCR)

Technology to convert scanned docs to text

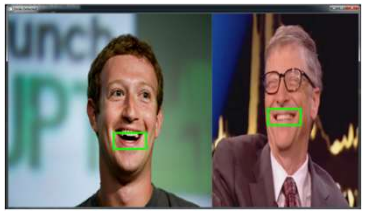
- If you have a scanner, it probably came with OCR software



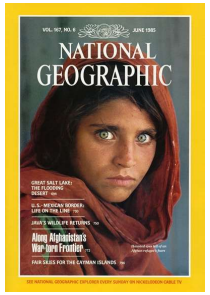
Face detection



Smile detection?

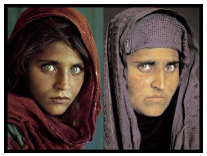


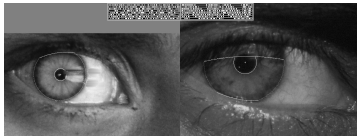
Face recognition



Who is she?

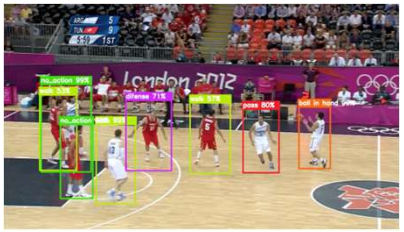
Vision-based biometrics





Special effects: shape capture

Sports



Smart cars

Robotics

Medical imaging

3D imaging MRI, CT

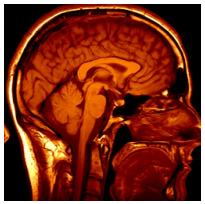


Image guided surgery
Grimson et al., MIT

How it works ?

Computer Vision as ML Pipeline

Preprocessing

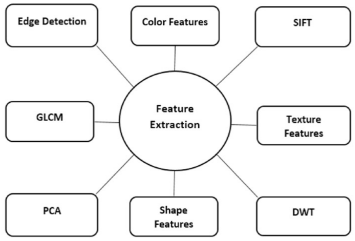
1.Pixel brightness transformations/ Brightness corrections
2.Geometric Transformations
3.Image Filtering and Segmentation
4.Fourier transform and Image restauration

Learning Algorithm

Supervised
- Classification
  - Naïve Bayes
  - SVM
  - Random Decision Forest
- Regression
  - Linear
  - Logistic
  - Generalised

Unsupervised
- Clustering
  - K-means
  - KNN
- Dimensionality Reduction
  - PCA
  - SVD

Feature Engineering



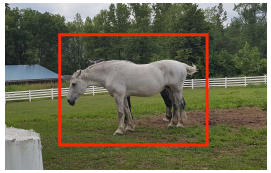
Computer Vision Trends

Accelerated by

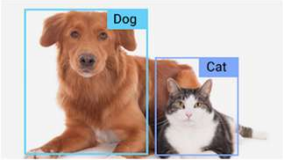
Deep Learning

Deep Learning models

CLASSFICATION



LOCALIZATION



OBJECT DETECTION

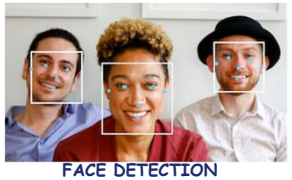
Deep Learning Models

SEGMENTATION

HAND LANDMARK DETETCION

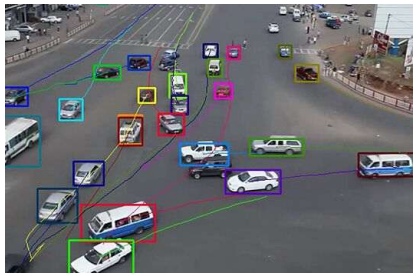
POSE ESTIMATION

Deep Learning Models



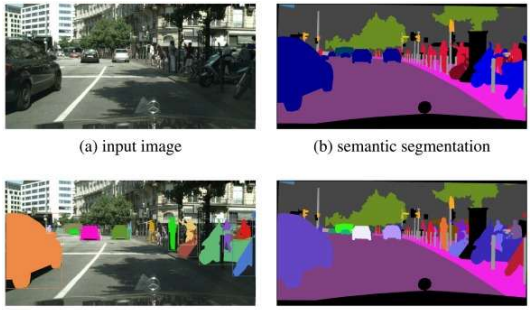
FACE DETECTION

FACE LANDMARK DETECTION



OBJECT TRACKING

Deep Learning Models



(a) input image (b) semantic segmentation

(c) instance segmentation (d) panoptic segmentation

Deep Learning Models

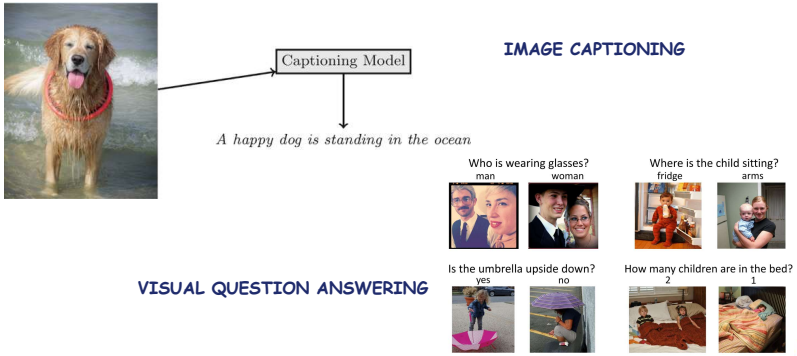
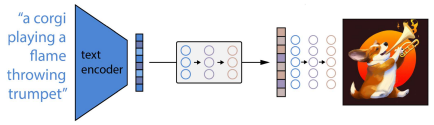


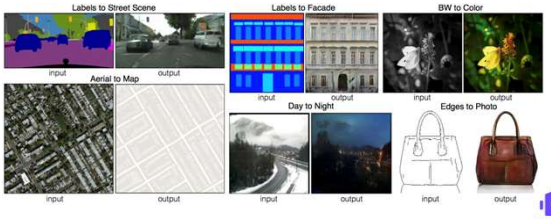
IMAGE CAPTIONING

VISUAL QUESTION ANSWERING

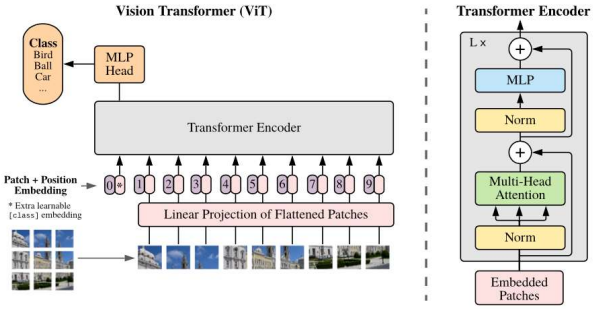
Deep Learning Models

GANs



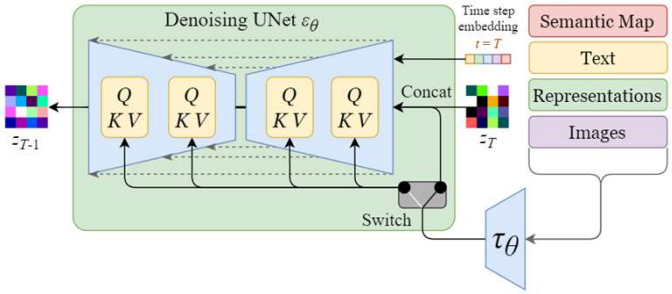


Deep Learning Models



VISION TRANSFORMERS

Deep Learning Models



DIFFUSION NETWORKS